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Q1. Smart Hospital Patient Registration System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        ArrayList<String> patientIds = new ArrayList<>();
        HashMap<String, String[]> patientMap = new HashMap<>();
        HashMap<String, Integer> deptCount = new HashMap<>();

        int maxPatients = 100;
        boolean running = true;

        while (running) {
            int choice = Integer.parseInt(sc.nextLine().trim());

            switch (choice) {
                case 1: // Register Patient
                    if (patientIds.size() >= maxPatients) {
                        System.out.println("Maximum Patient Limit Reached.");
                        break;
                    }

                    String id = sc.nextLine().trim();
                    String name = sc.nextLine().trim();
                    String mobile = sc.nextLine().trim();
                    String dept = sc.nextLine().trim();
                    String doctor = sc.nextLine().trim();

                    if (patientMap.containsKey(id)) {
                        System.out.println("Duplicate Patient ID. Registration Failed.");
                        break;
                    }

                    if (!name.matches("[a-zA-Z ]+")) {
                        System.out.println("Invalid Name. Only Alphabets Allowed.");
                        break;
                    }

                    if (!mobile.matches("\\d{10}")) {
                        System.out.println("Invalid Mobile Number. Must be 10 digits.");
                        break;
                    }

                    // Proper case for name
                    name = toProperCase(name);
                    dept = dept.toUpperCase().trim();
                    doctor = doctor.trim();

                    String[] patientData = {id, name, mobile, dept, doctor};
                    patientIds.add(id);
                    patientMap.put(id, patientData);

                    deptCount.put(dept, deptCount.getOrDefault(dept, 0) + 1);

                    System.out.println("Patient Registered Successfully.");
                    break;

                case 2: // Search Patient
                    String searchId = sc.nextLine().trim();
                    if (patientMap.containsKey(searchId)) {
```

```

        String[] p = patientMap.get(searchId);
        System.out.println("Patient Details");
        System.out.println();
        System.out.println("ID : " + p[0]);
        System.out.println();
        System.out.println("Name : " + p[1]);
        System.out.println();
        System.out.println("Mobile : " + p[2]);
        System.out.println();
        System.out.println("Department : " + p[3]);
        System.out.println();
        System.out.println("Doctor : " + p[4]);
    } else {
        System.out.println("Patient Not Found");
    }
    break;

case 3: // Display All Patients
    System.out.println("-----");
    System.out.println("ID\tName\tDepartment");
    System.out.println("-----");
    for (String pid : patientIds) {
        String[] p = patientMap.get(pid);
        System.out.println(p[0] + " " + p[1] + " " + p[3]);
    }
    break;

case 4: // Department-wise Count
    System.out.println("Department Count");
    for (Map.Entry<String, Integer> entry : deptCount.entrySet()) {
        System.out.println(entry.getKey() + " : " + entry.getValue());
    }
    break;

case 5: // Exit
    running = false;
    break;

default:
    System.out.println("Invalid Choice.");
}
}
sc.close();
}

// Method to convert name to Proper Case
public static String toProperCase(String input) {
    input = input.trim().replaceAll("\\s+", " ");
    String[] words = input.split(" ");
    StringBuilder result = new StringBuilder();
    for (String word : words) {
        if (word.length() > 0) {
            result.append(Character.toUpperCase(word.charAt(0)))
                .append(word.substring(1).toLowerCase())
                .append(" ");
        }
    }
    return result.toString().trim();
}
}
}

```

INPUT

```

1
P101
Rahul Kumar
9876543210
Cardiology
Dr.Arun
1

```

```
P102
Priya
9876543211
Ortho
Dr.Meena
3
4
2
P101
5
```

OUTPUT

```
Patient Registered Successfully.
Patient Registered Successfully.
-----
ID      Name      Department
-----
P101  Rahul Kumar  CARDIOLOGY
P102  Priya  ORTHO
Department Count
CARDIOLOGY : 1
ORTHO : 1
Patient Details

ID : P101

Name : Rahul Kumar

Mobile : 9876543210

Department : CARDIOLOGY

Doctor : Dr.Arun
```

Q2. Online Library Book Management System

CODE

```
nithiasdliabfljalb
```

INPUT

```
1
B101
Java programming language
Heartbeat Schildt
Programming
McGrow Hill

1
B102
Data structures
Mark Allen
Computer Science
Pearson

3
4
5
2
B101
6
```

OUTPUT

```
(no output)
```

Q1. Smart Hospital Patient Registration System

— Not Submitted —

Q2. Online Library Book Management System

— Not Submitted —

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Q1. Smart Hospital Patient Registration System

— Not Submitted —

Q2. Online Library Book Management System

— Not Submitted —